

1 版本信息【Version Information】

版本【Version】	时间【Date】	说明【Description】
V1.0	2019-12-12	初始版本
V2.0	2020-02-22	3.x固件版本
V3.0	2021-05-17	修改锅盖标定功能相关的命令接口为新的SDK接口
V4.0	2021-05-17	适配ASIC相关的指令
V5.0	2021-12-03	调整去锅盖标定的流程。

2 概述【Overview】

Tiny红外模组集成到整机中后，需要锅盖标定才可获得较好的成像及测温效果。本文系统的介绍了锅盖标定实现流程。锅盖标定计算在模组固件中已经实现，上位机只需调用libiruvc库中的相关命令接口按照标定流程操作即可完成标定过程。

相关的命令用法参考SDK中doc文件夹下接口说明文档。

【When the user integrates the module into the whole machine, it will inevitably cause a certain temperature measurement deviation due to the changes of thermal distribution and optical structure (add window, etc.). Therefore, Secondary calibration& Lid pattern noise correction is required in the whole machine. The Lid pattern noise correction has been realized in module firmware, and the master terminal only needs to call the relevant command interface in libiruvc library to complete the calibration process.

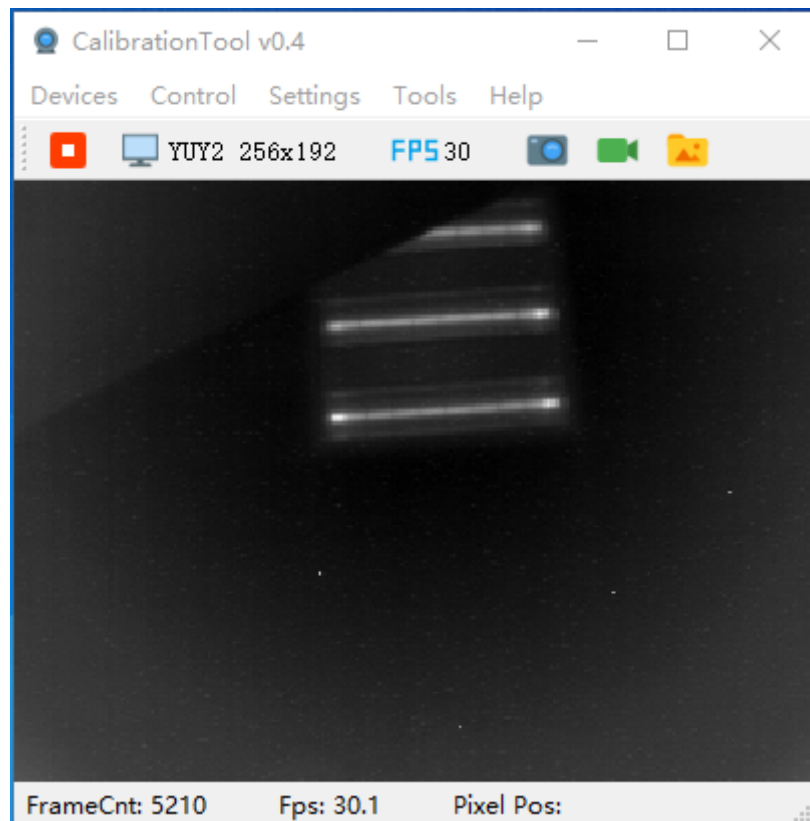
For related command usage, refer to the interface description document under the doc folder in the SDK.】

3 锅盖标定说明【Lid pattern noise correction Description】

3.1 Setp1

将模组安装好并确保当前模组达到热稳定状态(见图A，锅盖现象可能没那么严重)；

【Install the module and ensure that the module is thermally stable (see Figure A); 】



备注：在所有的标定之前(包含盲元标定，测温二次标定，锅盖标定)请先调用 `IRCMD` 类中 `restoreDefaultConfig` 接口的 `DEF_CFG_TPD` 方法，以重置标定数据，提高标定的准确性。

【Remarks: Before all calibrations (including blind element calibration, temperature measurement secondary calibration, pot lid calibration), please call the `DEF_CFG_TPD` method of the `restoreDefaultConfig` interface in the `IRCMD` class to reset the calibration data and improve the Calibration accuracy. 】

请注意：调用该方法之后，会重置所有的标定数据。【Note: After calling this method, all calibration data will be reset.】

即如果盲元标定结束之后，在锅盖标定的时候，调用了该方法，则盲元标定的数据会丢失。

【That is, if this method is called when the lid is calibrated after the blind element calibration is completed, the data of the blind element calibration will be lost.】

3.2 Setp2

调用 `setPropAutoShutterParameter` 函数的 `SHUTTER_PROP_SWITCH` 为0，关闭自动快门。

【Call `setPropAutoShutterParameter` function with `SHUTTER_PROP_SWITCH` 0 to close the auto FFC. 】

```

1  /**
2   * 设置自动快门参数【Set property page's auto shutter parameters】
3   * 需要 {@link #saveSpiConfig(int)} 的 {@link
CommonParams#SPI_MOD_CFG_PROPERTY_PAGE}才保存, 否则不保存
4   *
5   * @param shutter_param param during {@link CommonParams#SHUTTER_PROP_SWITCH
自动快门开关【Automatic shutter switch】},
6   *           {@link CommonParams#SHUTTER_PROP_MIN_INTERVAL 最小快
门间隔【Minimum shutter interval】},

```

```

7      *                               {@link CommonParams#SHUTTER_PROP_MAX_INTERVAL 最大快
      门间隔【Maximum shutter interval】}
8      *                               {@link
      CommonParams#SHUTTER_PROP_TEMP_THRESHOLD_OOC},
9      *                               {@link CommonParams#SHUTTER_PROP_TEMP_THRESHOLD_B 自
      动快门校正触发阈值【Automatic shutter correction trigger threshold】} ,
10     * @param value
11     * @return 0 is success
12     * @see #getPropAutoShutterParameter(int)
13     * @see #saveSpiConfig(int)
14     */
15     public int setPropAutoShutterParameter(int shutter_param, int value)

```

3.3 Setp3

调用 `shutterUpdate` 函数，进行快门校正将一温度均匀的物体对着模组(距离模组20mm左右)；

【Call `shutterUpdate` function to conduct FFC and an object with uniform temperature against the module (about 20mm away from the module); 】

```

1  /**
2   * 打快门【manual shutter】
3   * The firmware from 3.6 allow to shutter manually every 5 s
4   *
5   * @return 0 is success
6   */
7  public int shutterUpdate()

```

3.4 Setp4

调用 `rmCoverStsSwitch` 的函数，参数设置为 `RMCOVER_DIS`。关闭去锅盖功能。

【Call `rmCoverStsSwitch` function with parameter `RMCOVER_DIS` to disable remove cover function】

```

1  /**
2   * 去锅盖功能模块开关【Remove the lid function module switch】
3   * 立即生效，不可保存【Effective immediately, cannot be saved】
4   *
5   * @param status {@link CommonParams#RMCOVER_DIS},
6   *               {@link CommonParams#RMCOVER_EN}
7   * @return
8   */
9  public int rmCoverStsSwitch(int status)

```

3.5 Setp5

发送锅盖标定命令 `rmCoverAutoCalc`，`gain` 设置为1，3秒后即可拿开；

【Send Lid pattern noise correction command `rmCoverAutoCalc`；`gain` set to 1；3 seconds later, it can be removed; 】

```

1  /**
2      * 重新标定锅盖【Auto calculate the remove cover by firmware】
3      * 立即生效，不可保存【Effective immediately, cannot be saved】
4      *
5      * @param gain 1/2/4
6      * @return 0 is success
7      * @see #rmCoverStsSwitch(int)
8      */
9  public int rmCoverAutoCalc(int gain)

```

3.6 Setp6

调用 `rmCoverStsSwitch` 的函数，参数设置为 `RMCOVER_EN`。打开去锅盖功能。

【Call `rmCoverStsSwitch` function with parameter `RMCOVER_EN` to enable remove cover function】

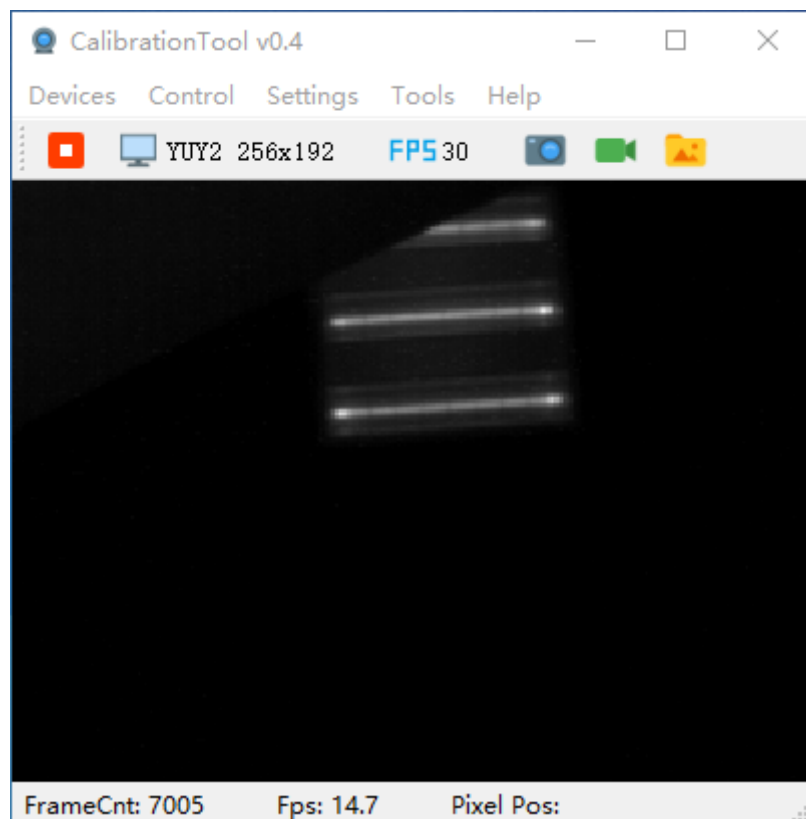
```

1  /**
2      * 去锅盖功能模块开关【Remove the lid function module switch】
3      * 立即生效，不可保存【Effective immediately, cannot be saved】
4      *
5      * @param status {@link CommonParams#RMCOVER_DIS},
6      *               {@link CommonParams#RMCOVER_EN}
7      * @return
8      */
9  public int rmCoverStsSwitch(int status)

```

去锅盖后的效果如图B；

【The effect is shown in figure B;】



3.7 Setp7

调用 `setPropAutoShutterParameter` 函数的 `SHUTTER_PROP_SWITCH` 为1，打开自动快门。

【Call `setPropAutoShutterParameter` function with `SHUTTER_PROP_SWITCH` as 1 to open the auto FFC.】

```
1  /**
2   * 设置自动快门参数【Set property page's auto shutter parameters】
3   * 需要 {@link #saveSpiConfig(int)} 的 {@link
4   CommonParams#SPI_MOD_CFG_PROPERTY_PAGE}才保存, 否则不保存
5   *
6   * @param shutter_param param during {@link CommonParams#SHUTTER_PROP_SWITCH
7   自动快门开关【Automatic shutter switch】},
8   *
9   *           {@link CommonParams#SHUTTER_PROP_MIN_INTERVAL 最小快
10  门间隔【Minimum shutter interval】},
11  *
12  *           {@link CommonParams#SHUTTER_PROP_MAX_INTERVAL 最大快
13  门间隔【Maximum shutter interval】}
14  *
15  *           {@link
16  CommonParams#SHUTTER_PROP_TEMP_THRESHOLD_OOC},
17  *
18  *           {@link CommonParams#SHUTTER_PROP_TEMP_THRESHOLD_B 自
19  动快门校正触发阈值【Automatic shutter correction trigger threshold】} ,
20  * @param value
21  * @return 0 is success
22  * @see #getPropAutoShutterParameter(int)
23  * @see #saveSpiConfig(int)
24  */
25  public int setPropAutoShutterParameter(int shutter_param, int value)
```

3.8 注意事项【Notes】

- 锅盖标定说明中的Setp7为非必须，如果标定前自动快门为关闭状态，这一部可以不执行，扔保持关闭状态；

【Setp7 in the calibration instructions of the pot lid is not necessary. If the automatic shutter is closed before calibration, this part can not be executed, and it will remain closed;】

如果标定前自动快门为打开状态，这一步可以执行，保持原状态。

【If the automatic shutter is open before calibration, this step can be performed, keeping the original state.】

- 标定失败后，再从Step1开始重新进行标准锅盖标定即可。

【After the calibration fails, restart the standard lid calibration from Step1.】

- 若模组为高低增益版本，需要分别在高低增益模式下标定锅盖形状。

【If the module is a high and low gain version, the shape of the pot lid needs to be calibrated in the high and low gain modes respectively.】

- 标定失败后，需要调用 `restoreDefaultConfig` 接口的 `DEF_CFG_TPD` 重置温度计算参数，重新进行二次标定

【If the calibration fails, call `DEF_CFG_TPD` of `restoreDefaultConfig` to restore the calibration data and perform secondary calibration again. 】

```
1  /**
2   * 恢复出厂设置,重置CFG参数【Restore factory settings, reset CFG
3   parameters】<br/>
```

```

3      * 立即生效, 可以保存【Effective immediately and can be saved】<br/>
4      * 重置后会停图, 需重新上电【Both resets will stop the map and require
re-powering】<br/>
5      *
6      * @param defaultCfgType {@link
CommonParams.DefaultConfigType#DEF_CFG_ALL} 重置所有参数【Reset all
parameters】,<br/>
7      *                                     {@link
CommonParams.DefaultConfigType#DEF_CFG_TECLSS_K},<br/>
8      *                                     {@link
CommonParams.DefaultConfigType#DEF_CFG_TPD} 重置测温参数【Reset
temperature measurement parameters】,<br/>
9      *                                     {@link
CommonParams.DefaultConfigType#DEF_CFG_PROP_PAGE} 重置默认算法参数 Reset
default algorithm parameters】
10     *                                     重置图像、快门参数【Reset image and shutter
parameters】
11     *                                     <br/>
12     *                                     {@link
CommonParams.DefaultConfigType#DEF_CFG_USER_CFG} ,<br/>
13     *                                     {@link
CommonParams.DefaultConfigType#DEF_CFG_DEAD_PIXEL} 重置盲元表【Reset the
blind meta table】<br/>
14     * @return see {@link IrcmdResult}
15     */
16     public int restoreDefaultConfig(CommonParams.DefaultConfigType
defaultCfgType)

```